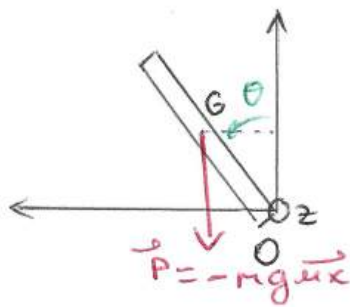


Chute chimisée



① Bilan forces:

- poids appliqué en G
- liaison pivot parfaite en O. $\vec{R}$

$$m_{Oz}(\vec{P}) = -\frac{D}{2} \sin \theta Mg$$

$$m_{Oz}(\vec{R}) = 0$$

$$L_{Oz} = \frac{1}{3} MD^2 \dot{\theta}$$

TMC / Oz à la chimisée de la ref terrestre :

$$\frac{dL_{Oz}}{dt} = \frac{1}{3} MD^2 \ddot{\theta} = \sum_i m_{Oz}(\vec{F}_i) = m_{Oz}(\vec{P}) + 0$$

$$\frac{1}{3} MD^2 \ddot{\theta} = -\frac{D}{2} \sin \theta \cdot Mg$$

$$\frac{1}{3} D \ddot{\theta} = -\frac{g \sin \theta}{2}$$

$$\ddot{\theta} = -\frac{3g \sin \theta}{2D}$$

② $\frac{dE_k}{dt} = \sum \mathcal{P}(\vec{F}_i)$

$$\mathcal{P}(\vec{P}) = m_{Oz}(\vec{P}) \cdot \dot{\theta} = -\frac{D}{2} \dot{\theta} \sin \theta Mg$$

$$\mathcal{P}(\vec{R}) = 0$$

$$\frac{d}{dt} \left(\frac{1}{2} J \dot{\theta}^2 \right) = -\frac{D}{2} \dot{\theta} \sin \theta \cdot Mg$$

$$J \ddot{\theta} = -\frac{D}{2} Mg \sin \theta$$

$$E_k = \frac{1}{2} J \dot{\theta}^2$$

$$\frac{1}{3} MD^2 \ddot{\theta} = -\frac{D}{2} Mg \sin \theta \quad \ddot{\theta} \neq \text{fonction nulle}$$

$$\ddot{\theta} = -\frac{3g \sin \theta}{2D}$$

③ PFD: $\Sigma \vec{F} = M \vec{a}_G$

$\vec{R} + \vec{P} = M \vec{a}_G$

En projetant suivant le vecteur $\vec{u}$:

$R_u + P_u = M \left(-\frac{D}{2} \ddot{\theta} \right)$

$P_u = -Mg \cos \theta$

$R_u = -\frac{D}{2} M \ddot{\theta}^2 + Mg \cos \theta$

$\vec{r}_G = \frac{D}{2} \vec{\theta} \times \vec{u}$

$\vec{a}_G = \frac{D}{2} \ddot{\theta} \vec{u} + \frac{D}{2} \dot{\theta} \frac{d\vec{u}}{dt}$

$= \frac{D}{2} \ddot{\theta} \vec{u} + \frac{D}{2} \dot{\theta} (-\dot{\theta} \vec{u})$

$= \frac{D}{2} \ddot{\theta} \vec{u} - \frac{D}{2} \dot{\theta}^2 \vec{u}$

④ TEC

$\Delta E_k_{t=0 \rightarrow t} = \Sigma W(\vec{F})$

$\frac{1}{2} J \dot{\theta}^2 - 0 = \frac{MgD}{2} (1 - \cos \theta)$

$\dot{\theta}^2 = \frac{MgD}{J} (1 - \cos \theta)$

$= \frac{MgD}{\frac{1}{3} M D^2} (1 - \cos \theta)$

$\dot{\theta}^2 = \frac{3g}{D} (1 - \cos \theta)$

$W(\vec{F})_{0 \rightarrow t} = \int_0^t \mathcal{P}(\vec{F}) dt$

$W(\vec{P})_{0 \rightarrow t} = 0$

$W(\vec{F}) = \int_0^t \frac{D}{2} \dot{\theta} \sin \theta \cdot Mg dt$

$= \frac{MgD}{2} \int_0^t -\cos \theta dt$

$= \frac{MgD}{2} [-\cos \theta]_0^t$

$= \frac{MgD}{2} (1 - \cos \theta)$

⑤ $R_u = -\frac{D}{2} M \left(\frac{3g}{D} (1 - \cos \theta) \right) + Mg \cos \theta$

$= -\frac{3}{2} gM + \frac{3}{2} gM \cos \theta + Mg \cos \theta$

$= -\frac{3}{2} gM + \frac{5}{2} gM \cos \theta = Mg \left(\frac{5}{2} \cos \theta - \frac{3}{2} \right)$

⑥ La cheminée décolle quand $R_u = 0$

$$\Leftrightarrow \frac{5}{2} \cos \theta = \frac{3}{2}$$

$$\Leftrightarrow \cos \theta = \frac{3}{5}$$

$$\Leftrightarrow \theta = \arccos\left(\frac{3}{5}\right)$$

$$\Leftrightarrow \boxed{\theta_d = 53^\circ}$$